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NIANZHAO ZHENG, Waseda University, Tokyo, Japan

JIALONG LI, Waseda University, Tokyo, Japan

NIANYU LI

MINGYUE ZHANG, Southwest University, Chongqing, China

JINYU CAI, Waseda University, Tokyo, Japan

KENJI TEI, Waseda University, Tokyo, Japan

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Exploring Optimal eHMI Display Location for Various Vehicle Types: A VR User Study

Nianzhao Zheng
Waseda University
Tokyo, Japan
protoss@akane.waseda.jp

Jialong Li*
Waseda University
Tokyo, Japan
lijialong@fuji.waseda.jp

Nianyu Li
ZGC National Laboratory
Beijing, China
li_nianyu@pku.edu.cn

Mingyue Zhang
Southwest University
Chongqing, China
myzhangswu@swu.edu.cn

Jinyu Cai
Waseda University
Tokyo, Japan
bluelink@toki.waseda.jp

Kenji Tei
Tokyo Institute of Technology
Tokyo, Japan
tei@c.titech.ac.jp

ABSTRACT

External human-machine interfaces (eHMI) are anticipated to enhance pedestrian interactions with automated vehicles (AVs) and increase social acceptance. While previous studies have focused on enhancing eHMI design by exploring various modalities and color preferences, identifying the optimal eHMI location on vehicles remains a significant challenge. Existing research has begun to explore different display locations for eHMI, but a thorough examination of the factors related to vehicle type, such as vehicle size, that influence pedestrian preferences is still lacking. Therefore, we conducted a Virtual Reality user study to assess pedestrian reactions to various AVs equipped with eHMI in different positions. Our findings reveal that larger vehicles are perceived as less safe by pedestrians, yet their increased visibility from a distance, due to longer light bands, impacts pedestrians' crossing decisions. Additionally, the location and height of the eHMI relative to eye level are crucial, with inappropriate positioning leading to diminished effectiveness.

CCS CONCEPTS

• **Human-centered computing → User studies**; *Empirical studies in HCI*.

KEYWORDS

eHMI; display location; vehicle type; vehicle-pedestrian interaction

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*Corresponding Author: Jialong Li

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1 INTRODUCTION

The emergence of AVs is anticipated to revolutionize traffic systems, enhancing traffic efficiency and reducing fatal accidents [19, 24, 37]. However, this technological shift presents a unique challenge: the absence of a human driver removes critical non-verbal communication cues like eye contact and hand gestures[20, 21, 27, 33, 34], which are instrumental for vulnerable road users (VRUs) such as pedestrians and cyclists [4, 32]. The lack of such interpersonal communication in AVs not only contributes to a sense of discomfort among VRUs [13, 30] but also indirectly increases the potential for accidents.

To mitigate communication issues between AVs and VRUs, eHMI have been extensively explored [1, 7, 8, 10, 14, 31]. eHMIs are designed to display essential information from AVs, such as driving mode, current actions, yielding intentions, and situational awareness. However, inappropriate eHMI designs could result in varied pedestrian perceptions [1, 7], potentially causing ambiguity and confusion among VRUs. The design of eHMIs, therefore, must take into account a range of factors, such as modality [31], display color [18], and message type [8], among others, to ensure clarity and effectiveness. Among these, the display location of eHMIs on AVs is a crucial factor that has already been explored in several studies. For instance, studies cited in [17, 22] have experimented with attaching eHMIs to various vehicle components, such as the grill or windshield.

While existing studies have made initial attempts to adjust the location of eHMIs on AVs, they have not thoroughly explored the criteria for determining the optimal location, nor have they discussed the factors affecting the optimal location. We argue that the ideal location for eHMIs is likely influenced by the type of vehicle. This is due to several factors: firstly, the type of vehicle influences people's psychological perceptions and judgments[6, 12, 29], with an example being pedestrians often feeling less safe around larger vehicles [26]. Secondly, different vehicle types vary in height, meaning that the same position relative to vehicle components on different vehicles (e.g., below the windshield) will result in different absolute display heights. In addition, this difference in absolute height also means the difference in height relative to a pedestrian's eye level, affecting pedestrians' gaze behavior. Consequently, we advocate for a more detailed investigation into how the relationship between the best eHMI location and vehicle type impacts both visibility and user perception.

Figure 1: This figure illustrates three example conditions from our VR user study, which evaluated pedestrian interactions with automated vehicles across nine conditions (3 vehicle types × 3 eHMI display locations). The conditions shown are: left - Taxi with eHMI on the bumper, middle - Pickup with eHMI at the bottom of the windshield, and right - Bus with eHMI at the top of the windshield. In the Taxi+bumper condition, the eHMI displays a yielding state, animating with light segments extending inwards, while the other two conditions, Pickup+bottom of the windshield and Bus+top of the windshield, show a non-yielding state with a steady glow.

To this end, we conducted a user study with 10 participants to explore the factors influencing pedestrians' perceptions and preferences for the display location of eHMI. The study was performed using a Virtual Reality (VR) environment, designed to simulate realistic scenarios. Specifically, participants encountered Automated Vehicles (AVs) varying across two manipulated factors, each with three levels: vehicle types and eHMI locations. A total of nine conditions were evaluated, with Figure 1 illustrating three conditions featuring different levels of these manipulated factors. Pedestrians were asked to indicate their willingness to cross using a 'feel-safe' button, and subjective opinions were collected through questionnaires to assess the effectiveness of the various conditions.

2 RELATED WORK

2.1 Design of eHMI

As a viable implementation for communicating the driving mode or intent of AVs, numerous studies have focused on designing better eHMI from various perspectives. Various concepts, grouped into visual, auditory, and physical categories based on their featured cues, have been presented and investigated [31]. Among these, visual interfaces are the most commonly used. Dey et al. [15] propose six eHMI concepts based on the principle of effectively communicating the intent of eHMI to multiple VRUs. These concepts achieve this purpose by directly communicating with each VRU in their space or using a timer to indicate when the vehicle intends to stop. de Clercq et al. [9] examined the effects of four different eHMI concepts on pedestrians' crossing decisions compared to situations with no eHMI. The results indicated that eHMI increased pedestrians' sense of safety, but non-textual eHMIs, such as LED strips, could cause confusion and require learning. Some studies aim to provide knowledge for establishing standardization in eHMI design, similar to traditional vehicle lights like brake lights. Faas and Baumann [18] compared the effects of using white and turquoise as eHMI colors. They concluded that turquoise is a suitable color as it is novel and has no specific meaning in current traffic systems. Dey et al. [14] conducted an online survey to determine user preferences for the color and animation patterns of a light-band eHMI. While animation patterns had minimal impact, the study found that cyan might be more suitable for communicating vehicles' yielding intentions, which is the color commonly used now for light-based eHMI.

2.2 Display Location of eHMI

Various display locations for external communication concepts have been explored in prior research. While communication could occur on personal devices or street infrastructure [31], interfaces on vehicles are most commonly used. Eisma et al. [17] conducted a video-based experiment with an eye-tracking device to compare the performance of five locations proposed in previous studies. They suggested that eHMIs on the grill, windshield, and top of the vehicle are subjectively considered better for pedestrians to see clearly, thereby improving their decision-making efficiency. Guo et al. [22] employed animated clips with regular AVs equipped with six different modalities of eHMI and three different display locations. They found that modalities that are easy to understand were most effective at reducing decision time, while the location did not have a significant effect. Dey et al. [16] conducted an on-road test using an eye-tracking device to analyze pedestrians' gaze patterns. They observed pedestrians' crossing willingness when encountering a yielding vehicle. Results showed that the part of the vehicle pedestrians looked at shifted upwards gradually as the distance between the vehicle and pedestrian decreased. They suggested that the display location for an eHMI depends on the distance between vehicles and pedestrians. Furthermore, different road users may perceive eHMI differently. Kim et al. [28] conducted a web-based survey to explore differences in display location preferences between pedestrians and drivers. They found that both groups prefer an eHMI in the middle of a vehicle, but their responses varied for the least preferred display location. Bazilinskyy et al. [2] conducted a video-based experiment to examine the effects of various factors, including the display location, on user interaction. Their findings

indicated that display locations on the roof and bumper were superior to the windshield. They argued that this is due to the eHMI on the windshield being less visible, as it was presented at an angle. In this study, we conducted a detailed investigation into the impact of eHMI on pedestrians' crossing decisions and their perceptions, considering various locations on different vehicle types. Specifically, factors such as vehicle size, position relative to vehicle components, and the height of eHMI location were taken into account.

3 USER STUDY

3.1 Research Question

Our study intend to answer the following research questions:

- *RQ1: What impact does the vehicle type have on pedestrians' crossing decisions and perceptions?*
- *RQ2a: Do pedestrians have a preference for certain vehicle components as the eHMI location? Does this preference vary across different vehicle types?*
- *RQ2b: Does the height of eHMI location relative to human eye level affect its performance?*

3.2 Method

Participants will stand on the sidewalk in a virtual environment and observe several AVs approaching from the left side of a two-lane, two-way urban road, using the Quest 2 head-mounted display. The road is approximately 100m long, with vehicles traveling at 50km/h, approaching from a crossroad about 70m to the left of the participant, then driving past the participant and disappearing at the road's end. The Unity game engine[25] models the scenario in this virtual environment. Three factors are manipulated in this experiment: First is the type of vehicle. We chose three types of vehicles commonly seen on urban roads: taxi (small), pickup truck (medium), and bus (large). The approximate dimensions of these vehicles are shown in Table 1.

The second factor is the display location of the light band, with three positions chosen for each type of vehicle: the top edge of the windshield, the bottom edge of the windshield, and the bumper, as suggested by Eisma et al. [17]. These positions are considered more noticeable for pedestrians and suitable for equipping a light band eHMI. The specific heights of the light bands from the ground are detailed in Table 1.

The third manipulated factor is the yielding behavior of the AVs. Cyan, which is recommended by many studies[3, 14, 36], is used as the color of the light band to indicate the AV's yielding intent. To prevent participants from overtrusting the vehicles' yielding behavior, non-yielding vehicles are also included in the experiment. These non-yielding vehicles will drive at 50 km/h, and their light bands will remain on without any animation pattern. In contrast, yielding vehicles will start braking at a distance of 45m from the participant and stop at a distance of 5m. The deceleration rate is set to $2.4m/s^2$, which approximates the normal braking rate for vehicles driving at 50km/h, as concluded in previous research [11]. After stopping for 3.5 seconds, these vehicles will resume driving and complete their trip. To signal the intent of yielding to pedestrians, adapted from previous research [23], we use a symmetrical animation where two light segments extend inwards several times until the vehicle decides to drive again.

The traffic flow includes a total of 54 vehicles, divided into 3 blocks (one for each vehicle type) with 18 vehicles in each block (3 display locations × 3 repetitions × 2 yielding behaviors). A new vehicle is generated only when the previous one has completed its trip and disappeared. The vehicles are generated in sequential order, from small to large and display locations from high to low. For yielding and non-yielding behaviors, two different fixed orders are alternated to increase ambiguity for participants, preventing them from developing expectations about the yielding behavior of approaching vehicles.

3.3 Measurements

To measure objective performance, a 'feel-safe' button is implemented. The participant's task is outlined as follows: (1) Press the button on the controller when you feel safe to cross the road, (2) Continue pressing the button as long as you feel safe, (3) Release the button when you no longer feel safe to cross. This method, adapted from previous studies [1][9], aims to minimize interference with the participants' task while continuously measuring their willingness to cross.

Furthermore, participants' subjective opinions are assessed using a questionnaire. All items in the questionnaire employ a seven-point Likert scale (1=Strongly disagree to 7=Strongly agree), covering aspects such as perceived safety ("I feel safe when encountering the AV equipped with eHMI"), visibility/noticeability ("I can easily and clearly see the displayed signals."), trust ("I think the AV is reliable."), and understandability ("The signals are clear and intuitive."). Participants are also asked to rank the display locations for each vehicle type according to their preference. Finally, participants are encouraged to provide comments or notes on the experiment as open feedback.

3.4 Participants

Ten participants, with an average height of 176.6 cm (2 participants between 160-170 cm, 5 between 170-180 cm, 2 between 180-190 cm, and 1 between 200-210 cm), aged between 20 to 27 years (Mean = 24.2, SD = 1.87), were involved in the experiment. All participants have normal mobility and eyesight, with seven of them wearing glasses during the experiment. Informed consent forms were distributed prior to their participation, and data were processed in accordance with the recommendations of the ethics board of the university where the researchers responsible for the study are based.

3.5 Procedure

Before starting the experiment, participants were briefed about the task and asked to perform a practice session to ensure they understood and could successfully carry out their tasks. Once prepared, participants were asked to put on the head-mounted VR display to begin the experiment.

Participants were required to continuously indicate their willingness to cross the road throughout the experiment. Each block lasted approximately 5 minutes. After completing a block, participants were asked to remove the head-mounted display and fill out a questionnaire to assess their experience. As the use of a head-mounted display can cause discomfort, which may affect participants differently, they were allowed to rest as needed between blocks to

Table 1: Vehicle dimensions and light band heights.

Vehicle Type	Dimensions (L × W × H)	Light Band Height from Ground (m)		
		Top of Windshield	*Bottom of Windshield*	*Bumper*
Taxi	5.4m × 1.7m × 1.5m	1.4m	1.1m	0.5m
Pickup Truck	5.5m × 1.9m × 2.1m	2.0m	1.6m	0.9m
Bus	15.0m × 3.2m × 4.0m	4.0m	2.1m	0.5m

mitigate the effects of discomfort. After completing all three blocks and the associated questionnaires, a brief interview was conducted to gather participants' opinions about the experiment.

4 RESULTS AND DISCUSSIONS

Figure 2 and Figure 3 show the percentage of participants feeling safe to cross relative to the distance between pedestrians and vehicles. Figure 2 illustrates the cases of non-yielding vehicles. The percentage of participants feeling safe continuously drops, reaching a minimum between 20m and 10m from the vehicle. Figure 3 depicts the cases of yielding vehicles. Notably, as the vehicle size increases, there is a more rapid or earlier decline in the percentage of participants feeling safe. Additionally, for larger vehicles, the percentage tends to rise earlier. The plots were similar across different display locations of the same vehicle type. The vertical reference lines in the figures indicate the distances where the vehicles started braking and the eHMI changed states. In all conditions, the percentage of participants feeling safe continuously decreases after the vehicles begin to yield. After reaching a certain distance, the percentages start to rise, reaching its maximum around 5 m, where the vehicles come to a full stop. Subsequently, the percentage drops to a minimum as the vehicle starts to drive again and passes the participant.

Table 2 presents descriptive statistics for all subjective ratings. The results indicate that neither the effect of display location ($F = 1.08, p = .343$) nor the effect of vehicle type ($F = 2.99, p = .056$) on safety were significant. However, a significant interaction effect was observed between vehicle type and display location ($F = 5.29, p < 0.001$). Specifically, for taxis, participants rated the bottom of the windshield worse than the other two display locations. For pickup trucks, the bumper was rated significantly higher than the bottom of the windshield. For buses, the top of the windshield received the lowest score. Similar results were obtained for the other three dimensions. While the main effects of display location and vehicle type were not significant, a significant interaction effect was found between these two factors. Participants' rating trends were also reflected in their preference rankings. Although some participants noted that an easy-to-see position might also be ambiguous, such as the bottom of the windshield, most based their ratings on visibility, as quoted from one participant: 'Because I think this display location is clearer to see, I rated it more safe, more trustworthy, and easier to understand.'

Table 3 shows how participants ranked the display locations of different vehicle types according to their preference. A lower mean rank indicates a more favorable position. For taxis, the bottom of the windshield was ranked less favorably compared to the other two locations. For pickup trucks, the bumper received a more favorable ranking than the other positions. For buses, the top of the windshield was least favored, whereas the bottom of the windshield was slightly more preferred than the bumper.

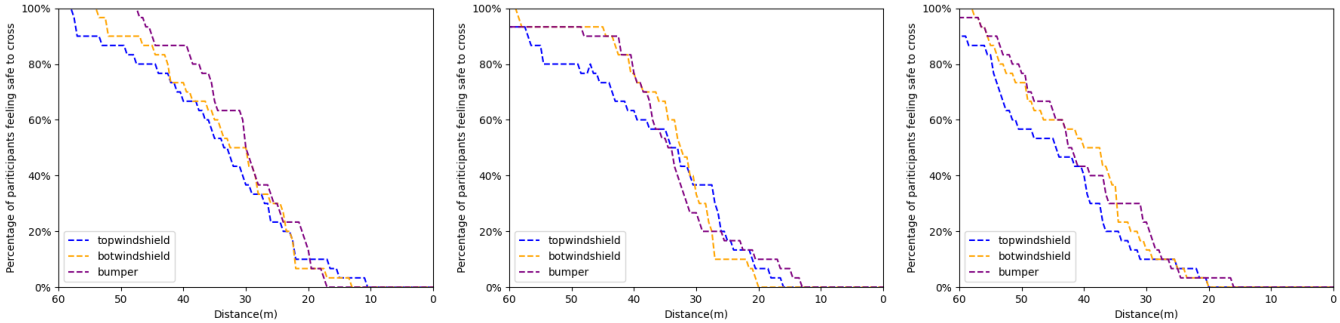
4.1 Discussions

4.1.1 RQ1: The Effect of Vehicle Type. In terms of vehicle size, participants generally felt less safe when encountering larger vehicles, aligning with findings from previous research [9, 26]. This is likely due to the longer braking distances associated with larger vehicles and their tendency to inflict greater damage in accidents. The button data indicated that participants' willingness to cross decreased faster and earlier when encountering pickups and buses compared to taxis. However, we also observed that while the percentage of participants feeling safe to cross in the presence of taxis began to rise around 25m to 20m, this increase occurred earlier for pickups and buses – around 30m to 25m for pickups and 40m to 35m for buses. During interviews, three participants noted this trend, explaining that larger vehicles, equipped with longer light bands due to their width, are easier to see from a longer distance. If pedestrians understand and trust the vehicles' yielding signals, the longer light band could enable them to make earlier crossing decisions.

Besides size, the vehicle's structure is another less-considered factor influencing perception. Most regular-sized passenger vehicles have a protruding front part for the engine and other devices, while buses or vans typically have a flatter front. Five participants mentioned that eHMIs attached to the protruding part are easier to see. Although the light band on the bumper of pickups and taxis did not show a notably better performance in influencing pedestrians' crossing decisions over other display locations, participants rated the bumper higher in subjective opinion surveys. The impact of such structural factors on pedestrians' perception requires further exploration in future research.

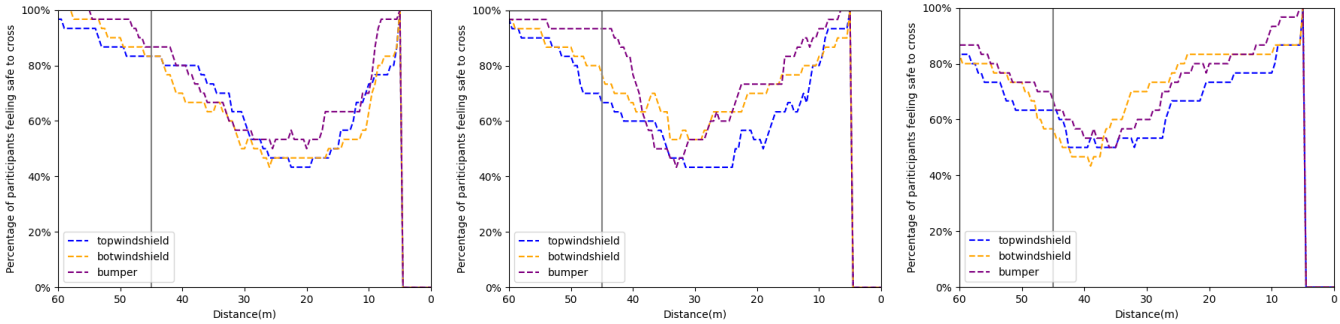
4.1.2 RQ2a: The Effect of Display Location Relative to Vehicle Components. Many participants reported their habitual gaze behaviors, with most tending to first look at the driver's cab or bumper upon noticing an approaching vehicle. While these habits influenced their choices to some extent, we found no clear preference for any specific vehicle component. However, participants' subjective opinions provided insights into what should be avoided when choosing the display location for a light band eHMI.

For taxis and pickups, six participants pointed out disadvantages of attaching the light band to the bottom of the windshield. The first issue is that this position intersects two colors: the color of the window and the color of the hood. Introducing a third color in such a position can make it difficult for pedestrians to distinguish. In addition, one participant mentioned that the degree of impact varies depending on the vehicle color. Existing research typically employs white or black vehicle models to carry the eHMI. However, this



(a) Non-yielding taxi (b) Non-yielding pickup (c) Non-yielding bus

Figure 2: Percentage of participants feeling safe to cross relative to the distance from non-yielding vehicles.



(a) Yielding taxi (b) Yielding pickup (c) Yielding bus

Figure 3: Percentage of participants feeling safe to cross relative to the distance from yielding vehicles.

Table 2: Descriptive statistics for subjective ratings.

Display location		Topwindshield	Bottomwindshield	Bumper
Safety	**Taxi**	5(1.78)	4.5(2.50)	4.8(1.51)
	Pickup	4.4(0.93)	3.9(1.43)	5.1(1.21)
	Bus	3(0.89)	5.3(0.23)	4.7(1.79)
Visibility/Noticeability	**Taxi**	4.8(1.51)	4(3.33)	4.9(1.21)
	Pickup	4.8(1.07)	3.5(2.72)	5.5(2.28)
	Bus	3.4(1.82)	5.6(0.27)	4.8(1.96)
Trust	**Taxi**	5.3(1.79)	4.3(2.46)	5.1(0.99)
	Pickup	4.7(1.12)	4(1.78)	5.3(1.12)
	Bus	3.5(1.39)	5.7(0.23)	5.1(1.66)
Understandability	**Taxi**	4.9(2.32)	4.3(2.23)	5(0.89)
	Pickup	5(1.11)	3.9(1.88)	5.6(0.49)
	Bus	3.7(2.01)	5.7(0.46)	5(1.56)

finding — specifically, the effects caused by the contrast between vehicle color and eHMI display color — remains unexplored and warrants further investigation in future studies. Secondly, signals or signs placed there might be misconstrued as decorative, leading pedestrians to overlook the vehicles' intentions. A similar issue was noted for the top of the windshield. Two participants expressed confusion, potentially mistaking the light band for a mark of a public vehicle, such as a police car. These scenarios should be considered while developing the standardization for eHMI design, as it is crucial to reduce ambiguity for pedestrians.

4.1.3 RQ2b: The Effect of Height Difference between eHMI and Human Eye Level. The impact of the display location, as mentioned above, is not pronounced in the case of buses. Additionally, the bottom of the windshield was rated as the best location for buses. We argue that this is due to the height difference between the pedestrian's eye level and the light band. Many participants noted

Table 3: Ranked preference on the display location of each vehicle type.

	Condition	Mean Rank	First Choice	Second Choice	Third Choice
Taxi	**Topwindshield**	1.8	3	6	1
	Bottomwindshield	2.5	2	1	7
	Bumper	1.7	5	3	2
Pickup	**Topwindshield**	2.2	1	6	3
	Bottomwindshield	2.4	2	2	6
	Bumper	1.4	7	2	1
Bus	**Topwindshield**	2.8	0	2	8
	Bottomwindshield	1.5	6	3	1
	Bumper	1.7	4	5	1

that one of their criteria for choosing a preferred display location is its suitability for their height or eye level. The results did not show a direct correlation that smaller height differences lead to higher ratings, but a significant height difference did result in lower ratings.

In the study by Eisma et al. [17], eHMI locations near the wheel or projections on the road, which might be too low for pedestrians, were not favored by participants. Similarly, in our study, the light band attached to the top of the windshield of a bus was almost unanimously considered an inappropriate display location, being too high for pedestrian viewing. One participant highlighted that the eHMI location also affects a pedestrian's vision. The eHMI should be attached at a position that allows pedestrians to easily see it while maintaining a clear view of the surrounding traffic information. The specific impact of the height difference between the eHMI and human eye level warrants further research.

5 CONCLUSION

This study conducted a VR user study to investigate the impact of various factors on pedestrians' perceptions of eHMI display locations. We found that larger vehicles generally make pedestrians more cautious when deciding to cross, while also enabling earlier decision-making due to a larger, more noticeable eHMI. Certain locations that may cause confusion for pedestrians were identified, suggesting a need to avoid these when placing eHMIs on specific vehicle types. Furthermore, our results suggest that the eHMI location should maintain a small height difference from human eye level and allow clear visibility of surrounding traffic information. Overall, our findings demonstrate that pedestrians' perceptions are significantly influenced by vehicle types, highlighting the complexity and necessity of designing and standardizing eHMIs with consideration for different vehicle types.

However, this study is only an initial step in exploring the optimal display location for eHMIs, and further research is needed to validate these findings. Future studies should include a broader range of vehicle types, considering factors such as vehicle color and shape, to gain a deeper understanding of the factors influencing the optimal eHMI location. Additionally, this study relied on a small, local student population; hence, a more diverse and representative sample is essential in future research. Additionally, the exploration of a broader array of scenarios, including complex maneuvers like turning or navigating parking spaces [5, 35], along with the conveyance of complicated information, remains a promising research avenue. Lastly, the VR environment used in the experiment may have affected participants' risk assessment, as indicated by the button data. A real-world study is therefore crucial to minimize the potential impact of the VR environment on cognitive ability and to verify the experiment's results.

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